

Loïc Chen

GAME ENGINE ARCHITECT & GRAPHICS PROGRAMMER

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Skills

- Languages: C++, Zig, JavaScript, Ruby, Python
- Graphics: Vulkan, Metal, OpenGL, WebGL, WebGPU
- Shading: GLSL, Metal Shading Language

Interests

Game engine programming, gameplay programming, physically-based rendering, GPU-driven pipelines, hybrid rendering, realtime VFX.

Professional Experience

Frontend Engineer · Mondraw Limited · Aug 2025 - Jun 2026

- Contributed to a cross-platform (iOS/Android) mobile poker game built on Cocos2d-x.
- Ported the game to Emscripten, redesigning the resource loading architecture and reducing runtime memory footprints by 70% so it could run smoothly on mobile browsers.

Graphics Engineer III · Phase GmbH · Feb 2023 - Mar 2024

- Contributed to an in-house high-performance 2D vector rendering engine (WebGL + WebAssembly) that renders complex Bézier spline graphics in realtime — the foundation of the team's UI/UX animation design product.
- Implemented Trim Path and Border Dash — two essential features required for vector-based animation.
- Diagnosed and fixed a class of Bézier curve rendering artifacts by solving edge cases in our implementation of Loop & Blinn (2006) curve categorization algorithm.

VR Gameplay Programmer · Lyra VR · Jun 2021 - Aug 2022

- Contributed to Galactic Catch, a VR fishing game (developed in Unity), shipped on the Meta Store and later won an Emmy Award.
- Built the customizable aquarium system: decoration placement and a dynamic fish-flocking behavior for various fishes.
- Implemented gameplay logic for cutscenes and tutorials.

Full-Stack Web Developer · Clubon · Dec 2019 - Mar 2021

- Sole programmer on a location-based ARG produced in promotional partnership with a Netflix series, working closely with game designers and artists on a deep and immersive narrative experience.
- Solo built PlayReal, a visual game editor (Ruby on Rails + React), not just enabling the team to deliver multiple games within two months, but also serving as our flagship product.
- Maintained our private server and supported legacy game updates.

Full-Stack Web Developer · Digital Learning Center @ National Taiwan University · Jun 2018 - Oct 2019

- Built a collaborative PDF annotation module (Ruby on Rails + React), for real-time markup and discussion between teachers and students.

Personal Projects

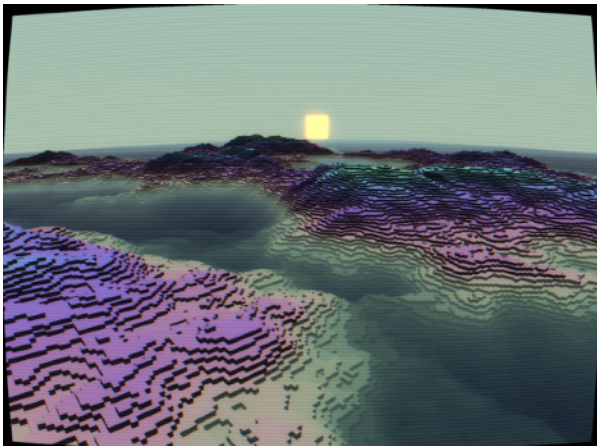


Project Vapor — A modern C++ 3D game framework with dual **Metal / Vulkan** backends.

[GitHub](#)

[Demo](#)

- Physically-Based Rendering based on **Disney BRDF**; **tiled light culling**; **raytraced hard shadows** for the main directional light; **glTF** model loading
- Physics integration (Jolt)
- UI framework integration (RmlUI)
- Audio engine with **3D audio** support
- Task scheduler integration (enkiTS)
- **ECS** gameplay architecture

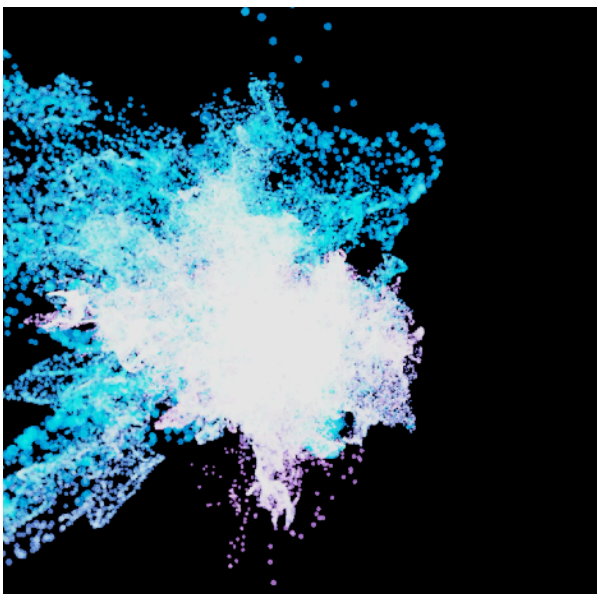


Atmospheric — A cross-platform C++ 2D / 3D game framework built on **OpenGL / WebGL**

[GitHub](#)

[Demo](#)

- PBR forward pipeline with a post-process stack (**physically based bloom**, tonemapping, color grading, chromatic aberration, etc.).
- Physics integration (Box2D / BulletPhysics)
- UI framework integration (RmlUI)
- Audio engine integration (Raudio)
- Simple **job system** — 3x loading speed of the single-threaded baseline
- Component-based gameplay architecture
- Lua scripting support; input buffering; heightmap terrain with **dynamic tessellation**



SDL3 GPU Experiment — My playground project for exploring **SDL3 GPU API**

[GitHub](#)

[Demo 1 \(GPU particles\)](#)

[Demo 2 \(GLTF\)](#)

- **glTF** model loading
 - **Compute-based particles** with mouse interaction. It can handle > 2 million particles on my Macbook Pro M3
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